



UNIVERSITATEA TEHNICĂ
DIN CLUJ-NAPOCA



Built Like a System.

Gym Membership Management System

E1 - Project Implementation Plan

Academic Year 2025-2026

1. Team Composition

Name	Role	Group	Main Responsibilities
Bogdan Dóra	Team Leader	30332/1	Coordinates the team, defines architecture, manages deliverables, organises technical meetings, and liaises with the supervising instructor.
Salajanu Mircea Vasile	Frontend Developer	30136/2	Designs and implements the graphical user interface (WinForms/WPF/WinUI), integrates UI components with application logic, validates user input.
Simionca Alexandru Petru Florin	Backend Developer	30136/2	Develops business logic components, manages database access via Entity Framework, implements services and APIs.
Kovacs Erik	Full-Stack Developer	30136/2	Contributes to both frontend and backend development, integrates system components and optimises application workflows.
Luidort Zsuzsa	Tester (QA Engineer)	30332/1	Defines test cases, creates unit tests, performs functional testing, identifies and reports software bugs.

2. Selected Project Topic

Gym Membership Management System

A desktop application developed in **C# (.NET)** for managing gym memberships, members, trainers, and subscription plans. The system targets gym administrators and staff who need an efficient, reliable tool to handle day-to-day operations.

3. Description of the Proposed Software System

3.1 Overview

The Gym Membership Management System is a desktop application written in **C# (.NET)**. The application is structured using a **3-tier architecture** - separating the user interface, the business logic, and the data access layer. We will apply SOLID principles and use the Repository Pattern to keep the code organized and easy to maintain. Input validation and business logic will be handled in the backend, and **database transactions** will be used where needed to keep data consistent (for example, when processing payments).

3.2 Key Features

Feature	Description
Member Management	Add, edit, delete and search members; store personal details and membership status. Includes filtering by status, plan type, and date range.
Subscription Plans	Define and manage subscription types (monthly, quarterly, annual) with pricing.
Trainer Management	Register trainers, assign specializations, and link them to members.
Attendance Tracking	Record and view gym check-ins per member.
Payments & Renewals	Track payment history and trigger membership renewal notifications. Payment processing uses database transactions to ensure data integrity.
Reports & Filtering	Generate summary reports; filter by status, date range, or plan type.
User Authentication	Secure login for admin/staff roles.

3.3 User Interface

The application provides a **graphical user interface** built with **WinForms/WPF/WinUI**, featuring a **main menu** and **at least 4 windows/pages** (Login, Dashboard, Member Management, Subscription Plans, Payments, Reports). UI controls used include **textboxes, comboboxes, checkboxes, radio buttons, and lists**. All user input is validated before being passed to the business logic layer.

3.4 Architecture

The application follows a **3-tier architecture**:

- **Presentation Layer:** WinForms/WPF/WinUI GUI – main menu, forms, and windows; UI controls for user interaction; input validation.
- **Business Logic Layer:** C# classes implementing SOLID principles, the Repository Design Pattern, backend validation, and transaction management.
- **Data Layer:** SQL Server database managed via **Entity Framework Core**

3.5 Database Schema (Main Tables)

- Members
- Subscriptions
- SubscriptionPlans
- Trainers
- Payments
- Attendance

4. Our Implementation Plan

Stage	Deadline	Activities & Deliverables
Stage 1 – Analysis	Week 5 (Sem. II)	Define system requirements; select project topic; define overall architecture of the system. Deliverable: E1 - Project Implementation Plan.
Stage 2 – Modeling	Week 10 (Sem. II)	Develop OOP model; design class structure and relationships; define application architecture; design GUI mockups; define test cases and validation scenarios. Deliverable: E2 - Software Modeling and Design Specifications.
Stage 3 – First Development Iteration	Week 14 (Sem. II)	Implement the software prototype; test implemented functionalities; present the project. Deliverables: E3.1 - Software Prototype (Release 1) + E3 - Final Report (project documentation).
Stage 4 – Future improvements	-	Improve the application; implement additional functionalities; optimize source code. Deliverable: E3.2 - Software Prototype (Release 2).

5. Task List

ID	Assigned To	Task Description	Timeline
T01	Bogdan Dora	Define system requirements and overall architecture	Wk 3-4
T02	Bogdan Dora	Draft & submit E1 - Project Implementation Plan	Wk 4-5
T03	Bogdan Dora	Read technical documentation and guides on multitier architecture, Repository Pattern, Entity Framework, and C# best practices to support implementation	Wk 5-8
T04	Bogdan Dora + assisted by the team	Design OOP model: class diagrams, use-case diagrams, architectural & functional design; define data persistence solutions; create class skeletons	Wk 6-8
T05	Bogdan Dora	Draft & submit E2 - Software Modeling and Design Specifications	Wk 8-10

T06	Bogdan Dora	Establish internal deadlines; ensure all team members have clearly defined weekly tasks and necessary materials	Ongoing
T07	Bogdan Dora	Organize weekly online team meetings to discuss project progress and contribute collectively to project documentation; perform final check & submit all reports	Weekly
T08	Bogdan Dora	Discuss & establish test cases and GUI test cases together with the Tester before implementation starts	Wk 7-9

T09	Bogdan Dora	Complete a Weekly Activity Report each week; submit 2 reports together every 2 weeks to the supervising instructor	Weekly / Submit Every 2 Wks
T10	Bogdan Dora	Coordinate integration; review deliverables; draft & submit E3.1 - Software Prototype (Release 1)	Wk 13-14
T11	Bogdan Dora	Draft & submit E3 - Final Report	Wk 13-14
T12	Bogdan Dora	Draft & submit E3.2 - Software Prototype Release 2 (if applicable)	Retake
T13	Salajanu Mircea Vasile	Design UI mockups and wireframes (assist Team Leader)	Wk 6-8
T14	Salajanu Mircea Vasile	Implement main application windows (WinForms/WPF/WinUI); follow modularity, code reuse & documentation practices	Wk 9-12
T15	Salajanu Mircea Vasile	Integrate UI with business logic layer; collaborate with Team Leader on component integration	Wk 12-13
T16	Salajanu Mircea Vasile	Contribute to project documentation during weekly team meetings	Ongoing
T17	Simionca Alexandru Petru Florin	Set up Entity Framework and database schema	Wk 7-9
T18	Simionca Alexandru Petru Florin	Implement CRUD operations for Members & Subscriptions; follow modularity, code reuse & documentation practices	Wk 9-11
T19	Simionca Alexandru Petru Florin	Implement Payment and Attendance modules; collaborate with Team Leader on component integration	Wk 11-13
T20	Simionca Alexandru Petru Florin	Contribute to project documentation during weekly team meetings	Ongoing

T21	Kovacs Erik	Implement Authentication module; follow modularity, code reuse & documentation practices	Wk 8-9
T22	Kovacs Erik	Implement Trainer Management module	Wk 9-12
T23	Kovacs Erik	Integrate system components and assist in testing; collaborate with Team Leader on component integration	Wk 12-14
T24	Kovacs Erik	Contribute to project documentation during weekly team meetings	Ongoing
T25	Luidort Zsuzsa	Study Software Testing concepts and modern testing methods	Wk 3-6
T26	Luidort Zsuzsa	Define unit test cases and GUI test cases together with Team Leader before implementation starts	Wk 7-9
T27	Luidort Zsuzsa	Implement unit tests for all classes	Wk 9-11
T28	Luidort Zsuzsa	Perform functional testing, identify and report software bugs; verify correctness of application workflows	Wk 12-14
T29	Luidort Zsuzsa	Contribute to project documentation during weekly team meetings	Ongoing

6. Activity Schedule

Activity	Wk 3	Wk 4	Wk 5	Wk 6	Wk 7	Wk 8	Wk 9	Wk 10	Wk 11	Wk 12	Wk 13	Wk 14
Requirements & Architecture												
E1 – Implementation Plan												
OOP & Design Modelling												
UI Mockups & Design												
Test Case Definition												
E2 – Design Specifications												
DB Schema & EF Setup												
Authentication Module												
UI Implementation												
CRUD – Members & Subs												
Trainer Management												
Payments & Attendance												
Unit Tests Implementation												
Integration & UI-Logic												
Functional Testing												
E3.1 – Prototype Release 1												
E3 – Final Report												

Deadlines: **Wk5** (E1), **Wk10** (E2), **Wk14** (E3.1+E3)

■ Active sprint / work period

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